Name = arayana khare

Sap id = 590042432

Batch = 15

Question 1: Reverse Each Word in a Sentence Without Changing the Word Order

1. Aim

To write a menu-driven C program using a separate user-defined function to reverse each word in a sentence without changing the order of the words.

2. Algorithm

1. Start the program.

2. Declare a character array to store the sentence.

3. Read the sentence using `fgets()`.

4. Call the `reverseEachWord()` function.

5. Scan the sentence character by character.

6. Identify each word using spaces as separators.

7. Reverse the characters of each word using a temporary variable.

8. Maintain the original order of the words.

9. Display the modified sentence.

10. Stop the program.

3. C Program

```
#include <stdio.h>
#include <string.h>

#define SIZE 200

void reverseEachWord(char str[]);

int main()
{
    char str[SIZE];

    printf("Enter a sentence: ");
    fgets(str, SIZE, stdin);
    str[strcspn(str, "\n")] = '\0';

    reverseEachWord(str);

    printf("Reversed sentence: %s\n", str);

    return 0;
}

void reverseEachWord(char str[])
{
    int i = 0, start, end;
    char temp;

    while (str[i] != '\0')
    {
        while (str[i] == ' ')
            i++;

        if (str[i] == '\0')
            break;

        start = i;
```

```
        while (str[i] != '\0' && str[i] != ' ')
            i++;

        end = i - 1;

        while (start < end)
        {
            temp = str[start];
            str[start] = str[end];
            str[end] = temp;

            start++;
            end--;
        }
    }
}
```

4. Explanation

- `fgets()` reads the complete sentence, including spaces.
- `strcspn()` removes the newline character from the input.
- The variable `start` stores the starting index of a word.
- The variable `end` stores the ending index of the word.
- A temporary variable swaps the first and last characters.
- The process continues until the characters meet in the middle.
- The same procedure is repeated for every word without changing the word order.

5. Sample Output

```
Enter a sentence: What is special?
Reversed sentence: tahW si ?laiceps
```

Question 2: Print the Initials of a Name

1. Aim

To write a C program using a separate user-defined function to print the initials of a person's full name.

2. Algorithm

1. Start the program.
2. Declare a character array to store the full name.
3. Read the full name using `fgets()`.
4. Call the `printInitials()` function.
5. Check each character of the name.
6. Identify the first character and the characters following spaces.
7. Convert each initial to uppercase using `toupper()`.
8. Print each initial followed by a full stop.
9. Stop the program.

3. C Program

```
#include <stdio.h>
```

```
#include <string.h>
#include <ctype.h>

#define SIZE 200

void printInitials(char name[]);

int main()
{
    char name[SIZE];

    printf("Enter full name: ");
    fgets(name, SIZE, stdin);
    name[strcspn(name, "\n")] = '\0';

    printf("Initials: ");
    printInitials(name);

    printf("\n");

    return 0;
}

void printInitials(char name[])
{
    int i, newWord = 1;

    for (i = 0; name[i] != '\0'; i++)
    {
        if (isspace((unsigned char)name[i]))
        {
            newWord = 1;
        }
        else if (newWord)
        {
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
        }
    }
}
```

4. Explanation

- `fgets()` reads the complete name, including spaces.

- The variable `newWord` indicates whether the current character begins a new word.

- `isspace()` checks whether the current character is a whitespace character.

- When a space is encountered, `newWord` is set to 1.

- The first character of each word is converted to uppercase using `toupper()`.

- Each initial is printed followed by a full stop.

5. Sample Output

```
Enter full name: Alexander Grahambell
Initials: A.G.
```

Another example:

```
Enter full name: Rahul Kumar Singh
Initials: R.K.S.
```

3. Combined Menu-Driven C Program

The following program performs both operations through a menu using separate user-defined functions.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define SIZE 200

void reverseEachWord(char str[]);
void printInitials(char name[]);

int main()
{
    char str[SIZE];
    int choice;
```

```
    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Reverse each word\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");

        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(str, SIZE, stdin);
                str[strcspn(str, "\n")] = '\0';

                reverseEachWord(str);

                printf("Reversed sentence: %s\n", str);
                break;

            case 2:
                printf("Enter full name: ");
                fgets(str, SIZE, stdin);
                str[strcspn(str, "\n")] = '\0';

                printf("Initials: ");
                printInitials(str);
                printf("\n");
                break;

            case 3:
                printf("Program exited.\n");
                break;

            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 3);

    return 0;
}
```

```
void reverseEachWord(char str[])
{
    int i = 0, start, end;
    char temp;

    while (str[i] != '\0')
    {
        while (str[i] == ' ')
            i++;

        if (str[i] == '\0')
            break;

        start = i;

        while (str[i] != '\0' && str[i] != ' ')
            i++;

        end = i - 1;

        while (start < end)
        {
            temp = str[start];
            str[start] = str[end];
            str[end] = temp;

            start++;
            end--;
        }
    }
}

void printInitials(char name[])
{
    int i, newWord = 1;

    for (i = 0; name[i] != '\0'; i++)
    {
        if (isspace((unsigned char)name[i]))
        {
            newWord = 1;
        }
        else if (newWord)
        {
```

```
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
        }
    }
}
```

4. Sample Output

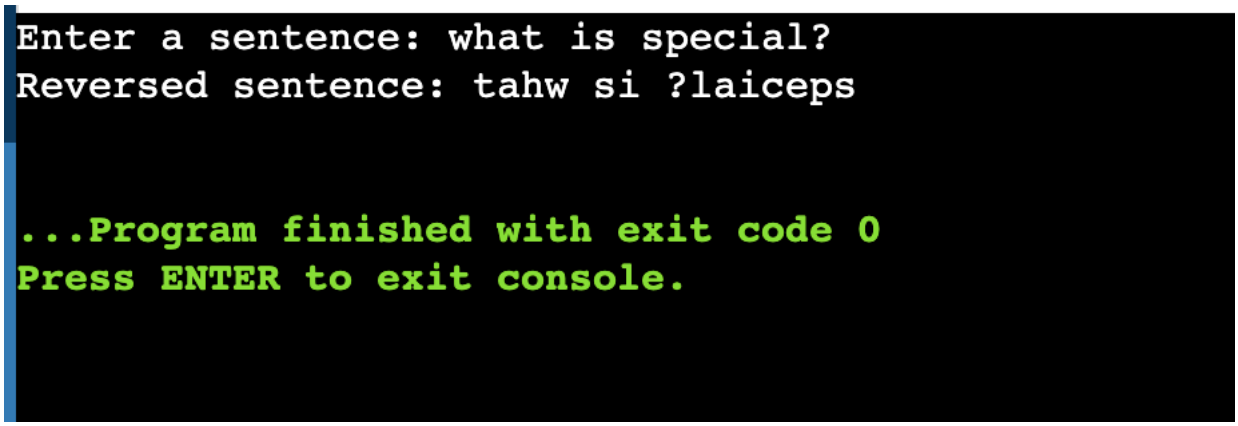
```
===== MENU =====
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: What is special?
Reversed sentence: tahW si ?laiceps

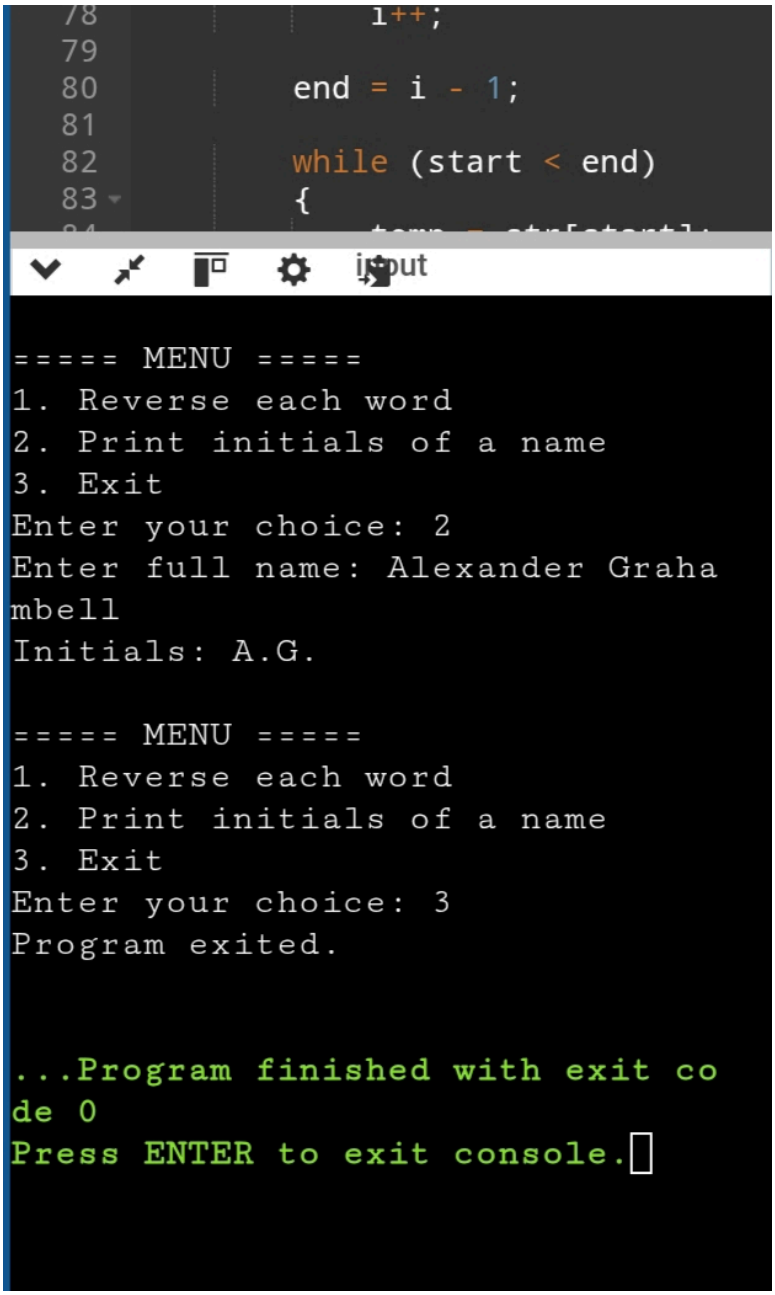
===== MENU =====
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter full name: Alexander Grahambell
Initials: A.G.

===== MENU =====
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 3
Program exited.
```

5. Result

The menu-driven C program successfully reverses each word in a sentence without changing the order of the words and prints the initials of a full name using separate user-defined functions.




i++;
end = i - 1;
while (start < end)
{
input
===== MENU =====
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter full name: Alexander Graha
mbell
Initials: A.G.
===== MENU =====
1. Reverse each word
2. Print initials of a name
3. Exit
Enter your choice: 3
Program exited.
...Program finished with exit co
de 0
Press ENTER to exit console.